

UNIVERSITY OF FRONTIER TECHNOLOGY, BANGLADESH

DEPARTMENT OF INTERNET OF THINGS AND ROBOTICS ENGINEERING (IRE)

FACULTY OF CYBER PHYSICAL SYSTEM ENGINEERING

LAB REPORT 01

DIGITAL WALLET BACKEND SYSTEM

(Implementation of Encapsulation and Polymorphism in Dart)

COURSE CODE: IRE 460

COURSE TITLE: MOBILE PLATFORM FOR IOT DEVICES SESSIONAL

SUBMITTED BY:

Wasif Azan Nafiz

Student ID: **2201045** | Session: 2022–2023

Department of Internet of Things and Robotics Engineering (IRE), UFTB

SUBMITTED TO:

Md. Rafiqul Islam

Lecturer

Department of Internet of Things and Robotics Engineering (IRE)

University of Frontier Technology, Bangladesh

NOVEMBER 03, 2026

1. Task Overview and Objectives

The task requires designing a lightweight backend architecture for a Digital Wallet application in Dart. The system must support two account categories: **Standard Accounts** and **Premium Accounts**. While standard accounts handle regular deposits, premium accounts offer incentive rewards by providing an automated cash-back on every deposit transaction.

2. Architectural Specifications

- **Encapsulation (`BankAccount`):**
  - Define a base class named `BankAccount`.
  - Encapsulate the balance field by making it private (`_currentBalance`).
  - Provide a public getter (`balance`) to access the funds without allowing external direct writes.
  - Create a `deposit(double amount)` method to handle account deposits.
- **Inheritance & Polymorphism (`PremiumAccount`):**
  - Create a subclass `PremiumAccount` inheriting from `BankAccount`.
  - Override the `deposit(double amount)` method so that premium users automatically earn a **5% cash-back reward** on their deposited amount.

3. Technical Approach

- **Data Hiding:** In Dart, private members are declared with an underscore (_). By declaring `_currentBalance`, direct assignment from outside the class is prevented. Read-only access is provided by `double get balance => _currentBalance;`.
- **Inheritance Structure:** The `PremiumAccount` class extends `BankAccount`, utilizing inherited behaviors and attributes.
- **Method Overriding:** The `deposit` method is marked with `@override`. The extra cash-back reward is computed as:

  $$\text{Reward} = \text{amount} \times 0.05$$

  The updated deposit amount including reward is then forwarded to `super.deposit(amount + reward);`.

4. Complete Dart Source Code (digital_wallet.dart)

```dart
// Base BankAccount class
class BankAccount {
  double _currentBalance = 0.0; // private balance field

  // Getter to read balance safely
  double get balance => _currentBalance;

  void deposit(double amount) {
    if (amount > 0) {
      _currentBalance += amount;
      print("Successfully deposited: \$$amount | Total: \$$_currentBalance");
    } else {
      print("Error: Deposit amount must be greater than 0");
    }
  }
}

// Premium account subclass
class PremiumAccount extends BankAccount {
  @override
  void deposit(double amount) {
    if (amount > 0) {
      double reward = amount * 0.05; // 5% bonus reward
      print("Reward earned: \$$reward (5% cashback)");
      super.deposit(amount + reward);
    } else {
      print("Error: Deposit amount must be greater than 0");
    }
  }
}

void main() {
  print("--- Standard Account Demo ---");
  BankAccount standard = BankAccount();
  standard.deposit(500);
  print("Final Balance: \$${standard.balance}\n");

  print("--- Premium Account Demo ---");
  PremiumAccount premium = PremiumAccount();
  premium.deposit(500);
  print("Final Balance: \$${premium.balance}");
}
```

5. Program Execution and Output

The Dart application was executed using the Dart runtime, producing the following console output:

```
--- Standard Account Demo ---
Successfully deposited: $500.0 | Total: $500.0
Final Balance: $500.0

--- Premium Account Demo ---
Reward earned: $25.0 (5% cashback)
Successfully deposited: $525.0 | Total: $525.0
Final Balance: $525.0
```

6. Verification and Summary

- **Standard Account Verification:** Depositing $500.0 into a standard account successfully updates the private balance from $0.0 to $500.0. The value is accessed safely via the getter.
- **Premium Account Verification:** Depositing $500.0 into a premium account triggers the overridden method, which calculates $25.0 as the 5% reward. The total amount credited becomes $525.0, verifying the cash-back logic.
- **Summary:** All requirements for encapsulation, inheritance, and method overriding were verified successfully with clean Dart code.